Chapter 5. Entire Functions

Jensen's formula, order of growth, infinite products, Weierstrass and Hadamard factorization

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Overview

Entire functions are holomorphic in the whole complex plane. The chapter is organized around three questions.

1. *Where can they vanish?* On any sequence without finite limit point (Weierstrass canonical factors, after Euler's product for $\sin \pi z$).
2. *How do they grow?* Faster growth allows more zeros; *Jensen's formula* links the zeros in a disc to the average of $\log|f|$ on its boundary.
3. *Are they determined by their zeros?* For finite order, up to a simple factor (*Hadamard factorization*).

1 Jensen's formula

Motivation. Jensen's formula lets growth control the number of zeros in a disc.

D_R and C_R denote the open disc and circle of radius R centered at 0. We exclude the function that vanishes identically.

Theorem 1.1 (Jensen's formula)

Let Ω be open with $\overline{D_R} \subset \Omega$, f holomorphic in Ω, $f(0) \neq 0$, $f \neq 0$ on C_R. If $z_1, \dots, z_N$ are the zeros of f in D_R (with multiplicities), then

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta. \tag{1}$$

Proof.

Step 1. If f_1, f_2 satisfy the hypotheses and (1), so does $f_1 f_2$, since $\log xy = \log x + \log y$ for $x, y > 0$ and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2.

Step 2. $g(z) = f(z)/((z - z_1)\cdots(z - z_N))$ is bounded near each z_j, hence extends holomorphically to Ω, nowhere zero on $\overline{D_R}$. By Step 1 it suffices to prove (1) for such g and for $z - z_j$.

Step 3. For such g: in a slightly larger disc $g = e^h$ (Theorem 6.2), so $\log|g| = \mathrm{Re}(h)$, and the mean value property (Corollary 7.3) gives $\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta$.

Step 4. For $f(z) = z - w$, $w \in D_R$: we need $\log|w| = \log\frac{|w|}{R} + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta$. As $\log|Re^{i\theta} - w| = \log R + \log|e^{i\theta} - w/R|$, it suffices that $\int_0^{2\pi} \log|e^{i\theta} - a|\, d\theta = 0$ for $|a| < 1$, i.e. $(\theta \mapsto -\theta)$ $\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0$. $F(z) = 1 - az$ has no zeros on the closed unit disc, so $\log|F| = \mathrm{Re}(G)$, $F = e^G$, is harmonic near it, and $\log|F(0)| = 0$; apply the mean value property. ∎

For f holomorphic on $\overline{D_R}$, let $n(r) = n_f(r)$ be the number of zeros of f (with multiplicities) in D_r, $0 < r < R$; $n(r)$ is non-decreasing. If $f(0) \neq 0$ and f does not vanish on C_R, then

$$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta - \log|f(0)|, \tag{2}$$

which follows from Jensen's formula and:

Lemma 1.2

If $z_1, \dots, z_N$ are the zeros of f inside the disc D_R, then

$$\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|.$$

Proof. $\sum_{k=1}^N \log\left|\frac{R}{z_k}\right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}$. With $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, $\sum_k \eta_k(r) = n(r)$, so $\sum_k \int_{|z_k|}^R \frac{dr}{r} = \sum_k \int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$. ■

2 Functions of finite order

Motivation. With growth of order ρ, Jensen's formula bounds the number of zeros.

Definition (order of growth)

An entire function f has an **order of growth** $\le \rho$ if $|f(z)| \le Ae^{B|z|^\rho}$ for all $z \in \mathbb{C}$, for some constants $A, B > 0$. The **order of growth** of f is $\rho_f = \inf \rho$ over all such $\rho > 0$.

For example, e^{z^2} has order of growth 2.

Theorem 2.1

If f is an entire function that has an order of growth $\le \rho$, then:
(i) $n(r) \le Cr^\rho$ for some $C > 0$ and all sufficiently large r.
(ii) If $z_1, z_2, \dots$ denote the zeros of f, with $z_k \neq 0$, then for all $s > \rho$ we have $\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. Assume $f(0) \neq 0$ (else use $F = f/z^\ell$, ℓ the order of the zero at 0: $n_f - n_F$ is constant and F has order $\le \rho$). By (2) with $R = 2r$,

$$\int_r^{2r} n(x)\frac{dx}{x} \le \int_0^{2r} n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{i\theta})|\,d\theta - \log|f(0)|.$$

Since n is increasing, the left side is $\ge n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$; and by the growth condition $\int_0^{2\pi}\log|f(2re^{i\theta})|\,d\theta \le \int_0^{2\pi}\log|Ae^{B(2r)^\rho}|\,d\theta \le C'r^\rho$ for large r. Hence $n(r) \le Cr^\rho$. For (ii),

$$\sum_{|z_k|\ge 1}|z_k|^{-s} = \sum_{j=0}^{\infty}\sum_{2^j\le|z_k|<2^{j+1}}|z_k|^{-s} \le \sum_{j=0}^{\infty} 2^{-js}n(2^{j+1}) \le c\sum_{j=0}^{\infty}2^{-js}2^{(j+1)\rho} \le c'\sum_{j=0}^{\infty}(2^{\rho-s})^j < \infty,$$

since $s > \rho$. ■

The condition $s > \rho$ cannot be improved:

Example 1

$f(z) = \sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$ satisfies $|f(z)| \le e^{\pi|z|}$, and taking $z = ix$ shows the order of growth is exactly 1. f vanishes to order 1 at each $n \in \mathbb{Z}$, and $\sum_{n\neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

Example 2

$f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty}(-1)^n \frac{z^n}{(2n)!}$ is entire with $|f(z)| \le e^{|z|^{1/2}}$, of order $1/2$. It vanishes at $z_n = ((n+1/2)\pi)^2$, and $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

Is there an entire f with zeros exactly at given $z_1, z_2, \ldots$? Necessarily $|z_k| \to \infty$ (Theorem 4.8); Weierstrass showed this suffices, inserting convergence factors into $(z-z_1)(z-z_2)\cdots$ without new zeros.

3 Infinite products

Motivation. Entire functions with infinitely many zeros need infinite products in place of $\prod(z-a_k)$.

3.1 Generalities

For complex a_n, the product $\prod_{n=1}^{\infty}(1+a_n)$ **converges** if $\lim_{N\to\infty}\prod_{n=1}^{N}(1+a_n)$ exists.

Proposition 3.1

If $\sum|a_n| < \infty$, then the product $\prod_{n=1}^{\infty}(1+a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.

Proof. Discarding finitely many terms, $|a_n| < 1/2$. With $b_n = \log(1+a_n)$ (power series (6) of Chapter 3), $\prod_{n=1}^{N}(1+a_n) = e^{B_N}$, $B_N = \sum_{n=1}^{N} b_n$. As $|\log(1+z)| \le 2|z|$ for $|z| < 1/2$, $|b_n| \le 2|a_n|$, so $B_N \to B$ and $e^{B_N} \to e^B$, which is $\neq 0$; with all $1+a_n \neq 0$ the full product is then non-zero. ■

Proposition 3.2

Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω. If there exist constants $c_n > 0$ such that $\sum c_n < \infty$ and $|F_n(z) - 1| \le c_n$ for all $z \in \Omega$, then:
(i) The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.
(ii) If $F_n(z)$ does not vanish for any n, then

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F_n'(z)}{F_n(z)}.$$

Proof. (i) Argue as in Proposition 3.1 with $|a_n(z)| \le c_n$; the estimates are uniform in z, and the uniform limit is holomorphic (Theorem 5.2).

(ii) Let $K \subset \Omega$ be compact and $G_N = \prod_{n=1}^{N} F_n$. Since $G_N \to F$ uniformly, Theorem 5.3 gives $G_N' \to F'$ uniformly on K. G_N is uniformly bounded below on K, so $G_N'/G_N \to F'/F$ uniformly on K; and $G_N'/G_N = \sum_{n=1}^{N} F_n'/F_n$ (Section 4). ■

3.2 Example: the product formula for the sine function

$$\sin \pi z = \pi z \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right) \tag{3}$$

for all $z \in \mathbb{C}$; it is derived from the sum formula for the cotangent

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\le N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \tag{4}$$

valid for $z \notin \mathbb{Z}$. The halves of $\sum 1/(z+n)$ with $n > 0$ and $n < 0$ diverge separately; only the symmetric interpretation converges.

Proof of (4). $F(z) = \pi \cot \pi z$ has the properties:

(i) $F(z+1) = F(z)$ for $z \notin \mathbb{Z}$;

(ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0;

(iii) F has simple poles at the integers and no other singularities. The function $\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ has the same three properties: for (i),

$$\sum_{|n|\le N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\le N} \frac{1}{z+n},$$

and let $N \to \infty$; parts (ii) and (iii) are evident from $\frac{1}{z} + \sum_{n\ge 1} \frac{2z}{z^2-n^2}$. So $\Delta(z) = F(z) - \sum \frac{1}{z+n}$ is 1-periodic with a removable singularity at 0 by (ii), hence at all integers, and is entire. It suffices to bound Δ in the strip $|\mathrm{Re}(z)| \le 1/2$; it is bounded for $|\mathrm{Im}(z)| \le 1$. For $z = x + iy$, $y > 1$,

$$\pi \cot \pi z = \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$$

is bounded, and

$$\Big|\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}\Big| = \Big|\frac{1}{x+iy} + \sum_{n=1}^{\infty} \frac{2(x+iy)}{x^2-y^2-n^2+2ixy}\Big| \le C + C\sum_{n=1}^{\infty} \frac{y}{y^2+n^2}.$$

The last sum is majorized by $\int_0^{\infty} \frac{y}{y^2+x^2}\,dx$ (decreasing integrand), independent of y. Similarly for $y < -1$. So Δ is bounded, constant by Liouville, and 0 as it is odd. ∎

Proof of (3). Let $G(z) = \sin \pi z$ and $P(z) = \pi z \prod_{n=1}^{\infty}(1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 3.2 shows P converges and, away from the integers, $\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$. Since $G'/G = \pi \cot \pi z$, (4) gives

$$\Big(\frac{P}{G}\Big)' = \frac{P}{G}\Big[\frac{P'}{P} - \frac{G'}{G}\Big] = 0,$$

so $P = cG$. Dividing by z and letting $z \to 0$, $c = 1$. ∎

4 Weierstrass infinite products

Motivation. Weierstrass's canonical factors make $\prod(1 - z/a_n)$ converge, so any zero sequence tending to infinity is attained.

Theorem 4.1 (Weierstrass)

For any sequence $\{a_n\}$ with $|a_n| \to \infty$ there is an entire f vanishing exactly at the a_n. Any other such function is $f(z)e^{g(z)}$ with g entire.

Repetitions are allowed in $\{a_n\}$, so this gives entire functions with prescribed zeros *and multiplicities* (the multiplicity of a zero a is the m with $f(z) = (z-a)^m g(z)$, $g(a) \neq 0$).

Proof. If f_1, f_2 both vanish exactly at the a_n, f_1/f_2 is entire and nowhere zero, so $f_1/f_2 = e^g$ (Theorem 6.2): the last statement.

To construct f: the naive product $\prod(1 - z/a_n)$ converges only for suitable $\{a_n\}$, so we insert exponential factors. Define the **canonical factors** of degree k:

$$E_0(z) = 1 - z, \qquad E_k(z) = (1-z)e^{z + z^2/2 + \cdots + z^k/k} \quad (k \ge 1).$$

Suppose f is to have a zero of order m at the origin and $a_1, a_2, \ldots$ are non-zero. The **Weierstrass product** is

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Fix $|z| < R$. The finitely many factors with $|a_n| \le 2R$ vanish exactly at the a_n with $|a_n| < R$. For $|a_n| > 2R$, $|z/a_n| \le 1/2$, and by Lemma 4.2 (c independent of n)

$$|1 - E_n(z/a_n)| \le c\Big|\frac{z}{a_n}\Big|^{n+1} \le \frac{c}{2^{n+1}}.$$

By Section 3, $\prod_{|a_n|>2R} E_n(z/a_n)$ is holomorphic and non-vanishing in $|z| < R$. As R is arbitrary, f has the required properties. ■

Lemma 4.2

If $|z| \le 1/2$, then $|1 - E_k(z)| \le c|z|^{k+1}$ for some $c > 0$.

Proof. For $|z| \le 1/2$, $1 - z = e^{\log(1-z)}$ (power series), so $E_k(z) = e^{\log(1-z)+z+z^2/2+\cdots+z^k/k} = e^w$ with $w = -\sum_{n=k+1}^{\infty} z^n/n$. Since $|z| \le 1/2$,

$$|w| \le |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \le |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \le 2|z|^{k+1}.$$

In particular $|w| \le 1$, so $|1 - E_k(z)| = |1 - e^w| \le c'|w| \le c|z|^{k+1}$. ■

Remark

The constant c is independent of k: one may take $c' = e$, $c = 2e$.

5 Hadamard's factorization theorem

Weierstrass writes $f = e^{g(z)} z^m \prod E_n(z/a_n)$. For finite order, Hadamard takes the canonical factors of constant degree and g a polynomial. Recall: order $\le \rho$ gives $n(r) \le Cr^\rho$ for large r and $\sum |a_n|^{-s} < \infty$ for $s > \rho$ (Theorem 2.1).

Theorem 5.1 (Hadamard)

Suppose f is entire and has growth order ρ_0. Let k be the integer so that $k \le \rho_0 < k + 1$. If $a_1, a_2, \ldots$ denote the (non-zero) zeros of f, then

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\le k$, and m is the order of the zero of f at $z = 0$.

Main lemmas.

Lemma 5.2

The canonical products satisfy

$$|E_k(z)| \ge e^{-c|z|^{k+1}} \text{ if } |z| \le 1/2 \quad \text{and} \quad |E_k(z)| \ge |1 - z|e^{-c'|z|^k} \text{ if } |z| \ge 1/2.$$

Proof. If $|z| \le 1/2$, $E_k(z) = e^{\log(1-z)+\sum_{n=1}^{k} z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w$, and $|e^w| \ge e^{-|w|}$ with $|w| \le c|z|^{k+1}$. If $|z| \ge 1/2$, $|E_k(z)| = |1-z||e^{z+z^2/2+\cdots+z^k/k}|$, and $|e^{z+\cdots+z^k/k}| \ge e^{-|z+\cdots+z^k/k|} \ge e^{-c'|z|^k}$ (as $|z| \ge 1/2$). ∎

The key is a lower bound for the product of canonical factors away from the zeros.

Lemma 5.3

For any s with $\rho_0 < s < k+1$, we have

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \ge e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \ldots$.

Proof. Write $\prod_n E_k(z/a_n) = \prod_{|a_n|\le 2|z|} E_k(z/a_n) \prod_{|a_n|>2|z|} E_k(z/a_n)$. For the second product, with no restriction on z, Lemma 5.2 gives

$$\Big|\prod_{|a_n|>2|z|} E_k(z/a_n)\Big| \ge \prod_{|a_n|>2|z|} e^{-c|z/a_n|^{k+1}} = e^{-c|z|^{k+1}\sum_{|a_n|>2|z|}|a_n|^{-k-1}},$$

and since $|a_n| > 2|z|$ and $s < k+1$, $|a_n|^{-k-1} = |a_n|^{-s}|a_n|^{s-k-1} \le C|a_n|^{-s}|z|^{s-k-1}$; as $\sum|a_n|^{-s} < \infty$, this product is $\ge e^{-c|z|^s}$. For the first product, by the second part of Lemma 5.2,

$$\Big|\prod_{|a_n|\le 2|z|} E_k(z/a_n)\Big| \ge \prod_{|a_n|\le 2|z|} \Big|1-\frac{z}{a_n}\Big| \prod_{|a_n|\le 2|z|} e^{-c'|z/a_n|^k}. \tag{5}$$

Here $\prod_{|a_n|\le 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k\sum_{|a_n|\le 2|z|}|a_n|^{-k}}$ and $|a_n|^{-k} = |a_n|^{-s}|a_n|^{s-k} \le C|a_n|^{-s}|z|^{s-k}$, so this is $\ge e^{-c|z|^s}$. For the first factor of (5): if z is outside the disc of radius $|a_n|^{-k-1}$ about a_n, $|a_n - z| \ge |a_n|^{-k-1}$, so

$$\prod_{|a_n|\le 2|z|} \Big|1-\frac{z}{a_n}\Big| = \prod_{|a_n|\le 2|z|} \Big|\frac{a_n-z}{a_n}\Big| \ge \prod_{|a_n|\le 2|z|} |a_n|^{-k-1}|a_n|^{-1} = \prod_{|a_n|\le 2|z|} |a_n|^{-k-2}.$$

Finally $\log\prod_{|a_n|\le 2|z|}|a_n|^{k+2} = (k+2)\sum_{|a_n|\le 2|z|}\log|a_n| \le (k+2)n(2|z|)\log 2|z| \le c|z|^s \log 2|z| \le c'|z|^{s'}$ for any $s' > s$, using $n(2|z|) \le c|z|^s$ (Theorem 2.1). Since $s > \rho_0$ was arbitrary, starting with s close enough to ρ_0 gives the lemma (with s' in place of s). ∎

Corollary 5.4

There exists a sequence of radii $r_1, r_2, \ldots$ with $r_m \to \infty$, such that

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \ge e^{-c|z|^s} \qquad \text{for } |z| = r_m.$$

Proof. Choose N with $\sum_{n\ge N}|a_n|^{-k-1} < 1/10$. For large integers L some $r \in [L, L+1]$ gives a circle missing the forbidden discs of Lemma 5.3: otherwise the intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$ would cover $[L, L+1]$, forcing $2\sum_{n\ge N}|a_n|^{-k-1} \ge 1$. Apply Lemma 5.3 with $|z| = r$. ∎

Proof of Hadamard's theorem. Let $E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n)$, entire as in Theorem 4.1: $|1 - E_k(z/a_n)| \le c|z/a_n|^{k+1}$ for large n (Lemma 4.2) and $\sum|a_n|^{-k-1} < \infty$ ($\rho_0 < s < k+1$). f/E is entire and nowhere zero, so $f/E = e^g$, g entire. By the order of f and Corollary 5.4,

$$e^{\mathrm{Re}(g(z))} = \Big|\frac{f(z)}{E(z)}\Big| \le c'e^{c|z|^s} \qquad \text{for } |z| = r_m,$$

so $\operatorname{Re}(g(z)) \le C|z|^s$ for $|z| = r_m$. Lemma 5.5 gives that g is a polynomial of degree $\le s$, hence $\le k$ (as $s < k+1$). ■

Lemma 5.5

Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies $u(z) \le Cr^s$ whenever $|z| = r$, for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\le s$.

Proof. Write $g(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 7.1,

$$\frac{1}{2\pi}\int_0^{2\pi} g(re^{i\theta})e^{-in\theta}\,d\theta = \begin{cases} a_n r^n & n \ge 0, \\ 0 & n < 0. \end{cases} \tag{6}$$

Taking complex conjugates, $\frac{1}{2\pi}\int_0^{2\pi} \overline{g(re^{i\theta})}e^{-in\theta}\,d\theta = 0$ for $n > 0$ (7). Since $2u = g + \overline{g}$, adding (6) and (7),

$$a_n r^n = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})e^{-in\theta}\,d\theta \quad (n > 0), \qquad 2\operatorname{Re}(a_0) = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})\,d\theta.$$

[Supplement]Since $\int_0^{2\pi} e^{-in\theta}d\theta = 0$ for $n > 0$, $a_n r^n = -\frac{1}{\pi}\int_0^{2\pi}[Cr^s - u(re^{i\theta})]e^{-in\theta}\,d\theta$, and $Cr^s - u \ge 0$ on the circles in question, so

$$|a_n| r^n \le \frac{1}{\pi}\int_0^{2\pi}[Cr^s - u(re^{i\theta})]\,d\theta = 2Cr^s - 2\operatorname{Re}(a_0).$$

Letting $r \to \infty$ along the sequence, $a_n = 0$ for $n > s$. ■

6 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 5 (book numbers kept).

Exercise 5.6 (Wallis's product)

(a) Prove Wallis's product formula

$$\frac{\pi}{2} = \prod_{m=1}^{\infty} \frac{2m \cdot 2m}{(2m-1)(2m+1)} = \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdots.$$

(b) Deduce that $\lim_{n\to\infty} \dfrac{2^{2n}(n!)^2}{(2n)!\,(2n+1)^{1/2}} = \sqrt{\pi/2}$.

Solution.

(a) Put $z = 1/2$ in the product formula (3) for the sine:

$$1 = \sin\frac{\pi}{2} = \frac{\pi}{2}\prod_{m=1}^{\infty}\Big(1 - \frac{1}{4m^2}\Big).$$

So the partial products $P_N = \prod_{m\le N}(1 - \frac{1}{4m^2})$ converge to $2/\pi \ne 0$. Hence $1/P_N \to \pi/2$, and

$$\frac{1}{P_N} = \prod_{m=1}^{N}\frac{4m^2}{4m^2-1} = \prod_{m=1}^{N}\frac{2m\cdot 2m}{(2m-1)(2m+1)}.$$

(b) Let $W_N = 1/P_N$.

Step 1. We have $\prod_{m\le N} 2m = 2^N N!$. The odd numbers below $2N$ and below $2N+2$ have products

$$\prod_{m\le N}(2m-1) = \frac{(2N)!}{2^N N!}, \qquad \prod_{m\le N}(2m+1) = \frac{(2N+1)!}{2^N N!}.$$

Step 2. Therefore

$$W_N = \frac{(2^N N!)^4}{(2N)!\,(2N+1)!} = \frac{2^{4N}(N!)^4}{((2N)!)^2(2N+1)} = \Big(\frac{2^{2N}(N!)^2}{(2N)!\,(2N+1)^{1/2}}\Big)^2.$$

Step 3. The bracket is positive and its square tends to $\pi/2$ by (a), so it tends to $\sqrt{\pi/2}$. ∎

Exercise 5.3 (Order of the theta function)

Fix τ with $t = \operatorname{Im}\tau > 0$. Show that the Jacobi theta function

$$\Theta(z\mid\tau) = \sum_{n=-\infty}^{\infty} e^{i\pi n^2\tau} e^{2\pi i n z}$$

is an entire function of z of order exactly 2.

Solution.

Step 1. (Upper bound.) The n-th term is $\le e^{-\pi n^2 t + 2\pi|n||z|}$. For $|n| \ge 4|z|/t$ this is $\le e^{-\pi n^2 t/2}$; for the at most $8|z|/t+1$ values $|n| < 4|z|/t$ it is $\le e^{8\pi|z|^2/t}$. Hence

$$|\Theta(z\mid\tau)| \le \Big(\frac{8|z|}{t}+1\Big)e^{8\pi|z|^2/t} + \sum_n e^{-\pi n^2 t/2} \le A e^{B|z|^2}.$$

The series converges uniformly on compact sets by the same bound, so Θ is entire (Theorem 5.2) of order ≤ 2.

Step 2. (Quasi-periodicity.) Replacing n by $n+1$ in the sum,

$$\Theta(z+\tau\mid\tau) = \sum_n e^{i\pi(n^2+2n)\tau} e^{2\pi i n z} = e^{-i\pi\tau} e^{-2\pi i z} \sum_n e^{i\pi(n+1)^2\tau} e^{2\pi i (n+1) z} = e^{-i\pi\tau - 2\pi i z}\,\Theta(z\mid\tau).$$

Step 3. ($\Theta \not\equiv 0$.) On the real line the series converges uniformly, so $\int_0^1 \Theta(x\mid\tau)\,dx = 1$ (only $n=0$ survives). Pick z_0 with $\Theta(z_0\mid\tau)\neq 0$ and write $y_0 = \operatorname{Im} z_0$.

Step 4. (Lower bound.) Let $z_m = z_0 + m\tau$. Since $|e^{-i\pi\tau - 2\pi i z}| = e^{\pi t + 2\pi \operatorname{Im} z}$ and $\operatorname{Im} z_j = y_0 + jt$, Step 2 gives

$$|\Theta(z_m\mid\tau)| = |\Theta(z_0\mid\tau)| \prod_{j=0}^{m-1} e^{\pi t + 2\pi(y_0+jt)} = |\Theta(z_0\mid\tau)|\, e^{\pi t m^2 + 2\pi y_0 m}.$$

If $|\Theta| \le A e^{B|z|^\rho}$ with $\rho < 2$, then, since $|z_m| \le |z_0| + m|\tau|$,

$$\pi t m^2 + 2\pi y_0 m + \log|\Theta(z_0\mid\tau)| \le \log A + B(|z_0| + m|\tau|)^\rho,$$

which fails for large m. Hence the order is exactly 2. ∎

Exercise 5.11 (Picard's little theorem for finite order)

Show that an entire function of finite order that omits two distinct values is constant.

Solution.

Step 1. If f has order $\leq \rho$ and omits $a \neq b$, $g = f - a$ is entire, nowhere zero, with $|g| \leq (A+|a|)e^{B|z|^\rho}$; by Hadamard (Theorem 5.1) $g = e^p$, p a polynomial.

Step 2. Since f omits b, $e^{p(z)} \neq b - a$ for all z. As $b - a \neq 0$, there is $w \in \mathbb{C}$ with $e^w = b - a$.

Step 3. If p were non-constant, some z has $p(z) = w$ (Corollary 4.6), so $e^{p(z)} = b-a$, a contradiction. So $f = a + e^p$ is constant. ■

Exercise 5.13 (Solutions of $e^z = z$)

Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$.

Solution.

Step 1. $f(z) = e^z - z$ satisfies $|f| \leq e^{|z|} + |z| \leq 2e^{|z|}$, so its order is $\rho_0 \leq 1$, and the integer k of Theorem 5.1 is 0 or 1. Also $f(0) = 1 \neq 0$.

Step 2. Suppose f had only finitely many zeros $a_1, \ldots, a_N$. By Hadamard's theorem, $f = e^P \prod_{n \leq N} E_k(z/a_n)$ with $\deg P \leq k \leq 1$. Each $E_k(z/a_n)$ is $(1 - z/a_n)$ times the exponential of a polynomial of degree $\leq k$. Collecting factors,

$$e^z - z = Q(z)e^{\alpha z+\beta}$$

for a non-zero polynomial Q and constants $\alpha, \beta \in \mathbb{C}$.

Step 3. ($x \to -\infty$.) $|Q(x)e^{\alpha x+\beta}| = |Q(x)|e^{x \operatorname{Re}\alpha+\operatorname{Re}\beta}$ while $|e^x - x| = |x| + o(1)$. $\operatorname{Re}\alpha > 0$ makes the left side $\to 0$, $\operatorname{Re}\alpha < 0$ makes it grow faster than any power; so $\operatorname{Re}\alpha = 0$.

Step 4. (Behaviour as $x \to +\infty$.) Then $|Q(x)e^{\alpha x+\beta}| = |Q(x)|e^{\operatorname{Re}\beta}$ grows at most polynomially. But $e^x - x \geq e^x/2$ for large x, a contradiction. So f has infinitely many zeros. ■